# Evaluation of Direct Oral Anticoagulant Use and Implementation of Pharmacist-led Direct Oral Anticoagulant Telephone Clinic in an Outpatient Federally Qualified Health Center

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## Background

Thromboembolic events such as cardioembolic stroke and venous thromboembolism (VTE), which includes deep vein thrombosis (DVT) and pulmonary embolism (PE), are major causes of morbidity and mortality. Stroke is the 5th leading cause of death in the United States, causing nearly 130,000 deaths each year. An estimated 15 percent of strokes are a result of untreated atrial fibrillation (A-fib). Atrial fibrillation is the most common type of heart arrhythmia and increases the risk of stroke five-fold.[1-3]

Warfarin has been the mainstay of oral anticoagulant treatment for VTE and atrial fibrillation since the 1950s. Despite its effectiveness in preventing thromboembolic events, warfarin therapy has several drawbacks including bleeding risk, drug interactions, dietary restrictions, and routine monitoring requirements.[4] Direct-Acting Oral Anticoagulants (DOACs) are a newer class of medications that may be more convenient and appropriate alternatives to warfarin. DOACs such as dabigatran, rivaroxaban, apixaban, and edoxaban were shown to be non-inferior to warfarin in randomized controlled trials.[5-7]

Although increasing evidence is showing that DOACs are a possible alternative therapy for prevention of thromboembolic events associated with VTE and atrial fibrillation, some prescribers are unaware of potential drug-drug interactions, drug-disease interactions, renal dose adjustments, appropriate durations of therapy, and cost implications. Pharmacists can play a large role in expanding the traditional anticoagulation ... encouraging ongoing adherence.[8-12]

## Purpose

The purpose of this study is to evaluate the prescribing patterns around DOACs in an outpatient setting and implement a pharmacist-led telephone clinic then collect data and compare to standard of care without pharmacist intervention. The pharmacist will then use evidence-based medicine to make appropriate recommendations regarding indications, interactions, adjustments, and durations of therapy.

Primary endpoint is hemorrhagic and/or thrombotic events. Secondary endpoints will be adherence and appropriateness of therapy based on the Adapted Medication Appropriateness Index (MAI).

## Health-Care Facility

Penobscot Community Health Care (PCHC) is by far the largest and most comprehensive of the 19 FQHC organizations in Maine and the 2nd largest of the 100 in the New England area. PCHC comprises of 16 practice sites with pharmacist-run clinics in 4 locations.

## Methods

**Adapted Medication Appropriateness Index (MAI)[11,12]**

| Criteria | Category | Instructions |
|---|---|---|
| Indication | A | Valid indication exists |
| | B | DOAC used as a last resort or indication does not fit within reimbursement criteria |
| | C | Off-label use |
| Choice | A | DOAC preferred choice: labile INR with VKA, CI to VKA, patient preference, recurrent stroke/VTE on VKA, resistance to VKA |
| | B | No contraindication, not yet tested, no recurrent stroke/VTE on VKA |
| | C | Not preferred choice: severe renal insufficiency, poor compliance, need for drug monitoring, severe hepatic impairment, recurrent VTE/stroke on current DOAC |
| Dosage | A | Receives daily dose as recommended |
| | C | Inappropriate daily dose (too low or too high) |
| Correct administration | A | Correct modalities of DOAC intake |
| | B | Limited clinical relevance for modalities of DOAC intake (timing of DOAC administration, DOAC given with or without food) |
| | C | Inappropriate modalities of DOAC intake (daily vs. twice daily) |
| Practical administration | A | No difficulties taking dosage form |
| | C | Difficulties taking the drug (BID dosing, not able to swallow capsules) |
| Drug-drug interactions | A | No DDI |
| | B | Potential DDI (caution or warning) without s/sx adverse event |
| | C | Contraindicated with disease, presents high risk, or DOAC used with caution and positive s/sx of worsening disease |
| Drug-disease interactions | A | No drug-disease interactions |
| | B | Potential interaction (caution or warning) without s/sx of adverse event |
| | C | Contraindicated with disease, presents high risk, or DOAC used with caution and positive s/sx of worsening disease |
| Therapeutic duplication | A | DOAC is the only antithrombotic |
| | B | Concomitant anticoagulation when switching therapy (DOAC to VKA) |
| | C | Duplication of antithrombotic |
| Duration of therapy | A | Duration in accordance with manufacturer indication |
| | C | Duration not appropriate based on recommendation |

**DOAC Utilization Screening**

- The electronic medical record will be used to identify patients who have received dabigatran, rivaroxaban, apixaban, or edoxaban as an active medication at PCHC.
- Inpatient and outpatient records will be reviewed to identify any adverse events such as hemorrhagic and/or thrombotic events, hospital admissions, or emergency room visits related to anticoagulant use.
- Adapted Medication Appropriateness Index (MAI) will be utilized to evaluate in appropriate prescribing.

**Patient enrollment**

- A pharmacist-led telephone clinic will be implemented for patients who are actively prescribed a DOAC and who provide consent.
- Patients will receive written consent form from the person during face-to-face appointment or in the mail.

**Pharmacist Intervention**

- Pharmacist will administer the organization's anticoagulation questionnaire over the phone to assess safety, efficacy, and adherence to patient's current DOAC therapy.
- Pharmacist will then use evidence-based medicine to make appropriate recommendations regarding indications, interactions, adjustments, and durations of therapy.
- Medication counseling and education will be provided regarding appropriate use and continued therapy.

**Patient Follow-Up**

- Pharmacist follow-up with patients once a month to monitor adherence, provide counseling, and assess if continued therapy is warranted (DVT/PE).
- Collect data and compare to standard of care without pharmacist intervention.

## Preliminary Results

| | # of patients | % of patients |
|---|---|---|
| Total patients evaluated | 89 | |
| # patients who met inappropriate indication (B, C) | 11 | 12.4 |
| # patients who met inappropriate choice (B, C) | 5 | 5.6 |
| # patients who met inappropriate dosage (C) | 6 | 6.7 |
| # patients who met inappropriate "correct" administration (B, C) | 16 | 18.0 |
| # patients who met inappropriate "practical" administration (C) | 2 | 2.2 |
| # patients who met inappropriate drug-drug interactions (A, B, C) | 23 | 25.8 |
| # patients who met inappropriate "drug-disease" interactions (B, C) | 2 | 2.2 |
| # patients who met inappropriate therapeutic duplication (B, C) | 1 | 1.1 |
| # patients who met inappropriate duration of therapy (C) | 6 | 6.7 |
| # patients who met at least 1 inappropriate criteria | 51 | 57.3 |

## Disclosure

Authors of this presentation have the following to disclose concerning possible financial or personal relationships with commercial entities that may have a direct or indirect interest in the subject matter of this presentation:

- Andrew Tran, PharmD: nothing to disclose

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